



The first extreme-loving organisms were discovered in Colorado, in Yellowstone National Park's sulfur-rich hot springs. In 1974, such organisms were classified as extremophiles.

STEPHEN HAWKING (b.1942)

Perhaps best known to the public for *A Brief History of Time* (1988), British theoretical physicist Stephen Hawking has made pioneering contributions to our understanding of black holes, cosmology, and quantum physics. He is almost completely paralyzed, a result of the degenerative condition known as motor neuron disease, with which he was diagnosed when he was just 21.



in the Afar Triangle, Ethiopia. The fossils were from a **3.2 million year-old *Australopithecus afarensis***—a species of bipedal (walked on two legs) hominid. The find was named "Lucy"—after the Beatles' song *Lucy in the Sky with Diamonds*, which was often played at the camp where the bones were found—and was by far the oldest such find at the time.

British theoretical physicist **Stephen Hawking** published a **revolutionary idea about black holes**. One of the findings of quantum physics is that empty space is constantly thronging with pairs of virtual particles, which exist only fleetingly, before

annihilating each other. Hawking uncovered a strange possibility: for any virtual pairs created at the event horizon (boundary) of a black hole, one member would disappear inside, while the other would travel out into space, the result being that black holes "radiate" particles. At first controversial, **Hawking radiation** is now part of mainstream physics, and the search is on to detect it. Also in physics, a rapid



Lucy
A reconstructed skull of an *Australopithecus afarensis*, a species of early bipedal apes.

succession of discoveries known as the **November Revolution**—which included the discoveries of the charm quark and the particle called J or psi—gave an enormous boost to the emerging understanding of **subatomic particles**, known as the Standard Model of particle physics (see panel, below).

IN JULY, AMERICAN

MICROBIOLOGIST Robert MacElroy coined the term "extremophiles" to refer to **organisms that thrive in extremes**—of acidity, pressure, or temperature, for example. This classification sparked interest from astrobiologists, since such organisms might be found in hostile environments on other planets, and from evolutionary biologists, as extremophiles are mostly primitive organisms that evolved early in Earth's history.

German geneticist **Rudolf Jaenisch** (b.1942) and American embryologist **Beatrice Mintz** (b.1921) reported that they had **created the first transgenic animal**, by introducing virus DNA into a mouse embryo. The pace of

development in genetics caused concern in the general public and the scientific community about the possible environmental effects of genetic engineering.

More concern about the environmental effects of scientific progress arose after scientists published their findings about the dangers of **chlorofluorocarbons (CFCs)**. These man-made compounds were widely used as refrigerants and propellents in aerosol cans. The researchers warned that **CFCs could deplete the ozone layer**, which protects the planet from harmful ultraviolet radiation.

In November, American anthropologist **Donald Johanson** (b.1943) discovered fossil fragments of a hominid skeleton

STANDARD MODEL OF PARTICLE PHYSICS

The "Standard Model," developed in the 1960s and 1970s, is still the best-fit explanation of fundamental particles—the most basic building blocks of matter and force. The model's success rests largely on its predictions of the existence of particles that have since been observed, including the charm quark in 1974. According to the theory, there are two families of fundamental particles: the fermions, which make matter, and the bosons, which carry force. The fermions are further divided into quarks—which normally occur in tightly bound twos or threes to make composite particles such as the proton and the neutron—and leptons, which occur singly and include the electron.

